PAPER 1

M1

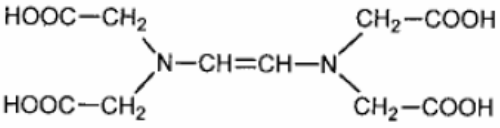
PART – I : CHEMISTRY

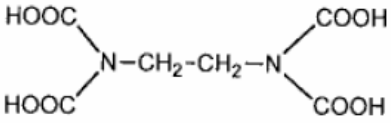
SECTION – I (Single Correct Choice Type)

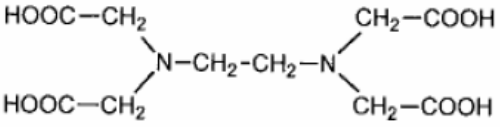
This Section contains **8 multiple choice questions**. Each question has four choices A), B), C) and D) out of which **ONLY ONE** is correct.

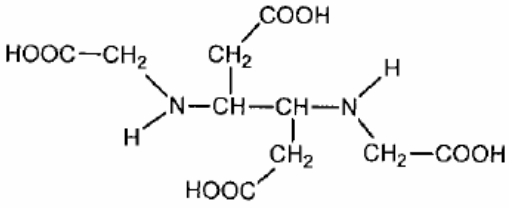
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1. The correct structure of ethylenediaminetetraacetic acid (EDTA) is

A) 

B) 

C) 

D) 

ANSWER: C

2. The ionization isomer of $[Cr(H_2O)_4Cl(NO_2)]Cl$ is

A) $[Cr(H_2O)_4(O_2N)]Cl_2$

B) $[Cr(H_2O)_4Cl_2](NO_2)$

C) $[Cr(H_2O)_4Cl(ONO)]Cl$

D) $[Cr(H_2O)_4Cl_2(NO_2)]\cdot H_2O$

ANSWER: B

3. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are

A) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3CH_2C\equiv CH$

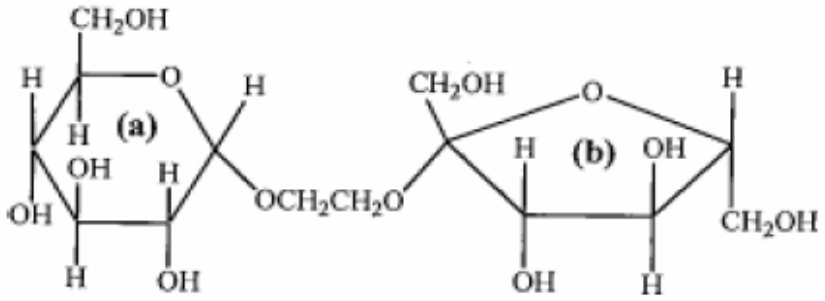
B) $BrCH_2CH_2CH_3$ and $CH_3CH_2CH_2C\equiv CH$

C) $BrCH_2CH_2CH_2CH_2CH_3$ and $CH_3C\equiv CH$

D) $BrCH_2CH_2CH_2CH_3$ and $CH_3CH_2C\equiv CH$

ANSWER: D

4. The correct statement about the following disaccharide is



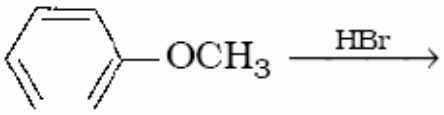
A) Ring **(a)** is pyranose with α-glycosidic link

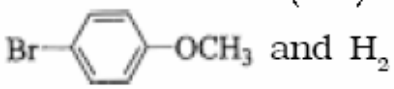
B) Ring **(a)** is furanose with α-glycosidic link

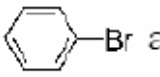
C) Ring **(b)** is furanose with α-glycosidic link

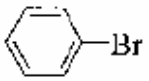
D) Ring **(b)** is pyranose with β-glycosidic link

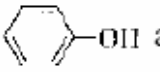
ANSWER: A

5. In the reaction  the products are

A)  and H_2

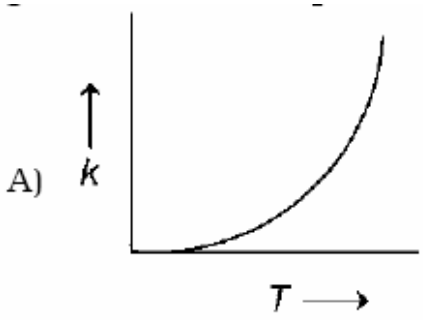
B)  and CH_3Br

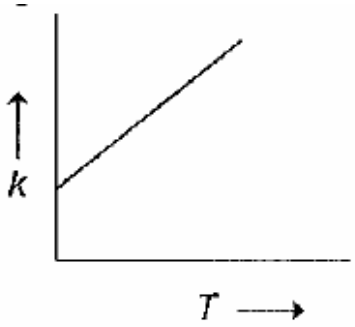
C)  and CH_3OH

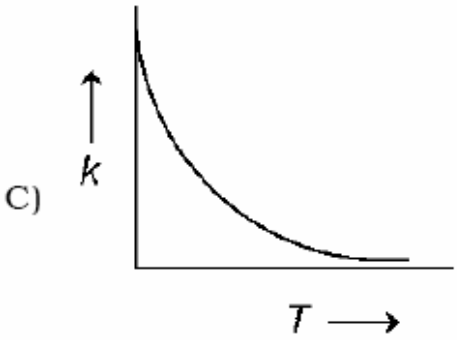
D)  and CH_3Br

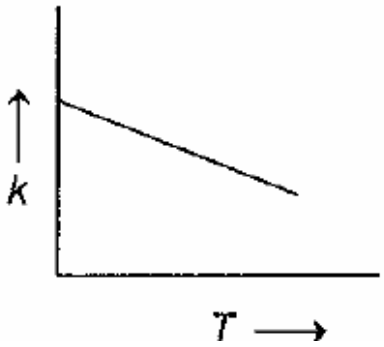
ANSWER: D

6. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is

A) 

B) 

C) 

D) 

ANSWER: A

7. The species which by definition has **ZERO** standard molar enthalpy of formation at 298 K is

A) Br_2 (g) B) Cl_2 (g) C) H_2O (g) D) CH_4 (g)

ANSWER: B

8. The bond energy (in **kcal mol^{-1}**) of a C–C single bond is approximately

A) 1 B) 10 C) 100 D) 1000

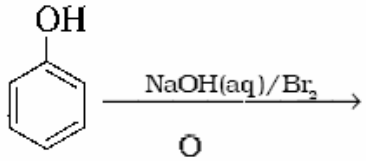
ANSWER: C

SECTION – II (Multiple Correct Choice Type)

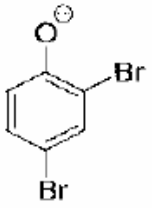
9. The reagent(s) used for softening the temporary hardness of water is(are)

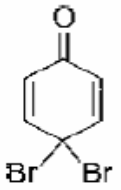
A) $Ca_3(PO_4)_2$ B) $Ca(OH)_2$ C) Na_2CO_3 D) NaOCl

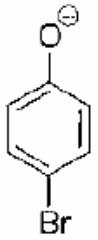
ANSWER: B

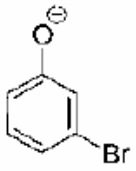
10. In the reaction 

the intermediate(s) is(are)

A) 

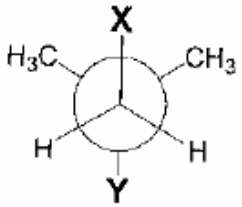
B) 

C) 

D) 

ANSWER: A and C

11. In the Newman projection for 2,2-dimethylbutane



X and **Y** can respectively be

A) H and H B) H and C_2H_5

C) C_2H_5 and H D) CH_3 and CH_3

ANSWER: B and D

12. Among the following, the intensive property is (properties are)

A) molar conductivity B) electromotive force

C) resistance D) heat capacity

ANSWER: A and B

13. Aqueous solutions of HNO_3, KOH, CH_3COOH, and CH_3COONa of identical concentrations are provided. The pair(s) of solutions which form a buffer upon mixing is(are)

A) HNO_3 and CH_3COOH
B) KOH and CH_3COONa
C) HNO_3 and CH_3COONa
D) CH_3COOH and CH_3COONa

ANSWER: C and D

SECTION – III (Paragraph Type)

Paragraph for Questions 14 to 16

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcanthite ($CuSO_4 \cdot 5H_2O$), atacamite ($Cu_2Cl(OH)_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($Cu_2(OH)_2CO_3$). However, 80% of the world copper production comes from the ore chalcopyrite ($CuFeS_2$). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction.

14. Partial roasting of chalcopyrite produces

A) Cu_2S and FeO
B) Cu_2O and FeO
C) CuS and Fe_2O_3
D) Cu_2O and Fe_2O_3

ANSWER: A

15. Iron is removed from chalcopyrite as

A) FeO
B) FeS
C) Fe_2O_3
D) $FeSiO_3$

ANSWER: D

16. In self-reduction, the reducing species is

A) S B) O^{2-}

C) S^{2-} D) SO_2

ANSWER: C

Paragraph for Questions 17 to 18

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is:

$M(s)|M^+(aq;\ 0.05\ molar)||M^+(aq;\ 1\ molar)|M(s)$

For the above electrolytic cell the magnitude of the cell potential $|E_{cell}| = 70$ mV.

17. For the above cell

A) $E_{cell} < 0;\ \Delta G > 0$ B) $E_{cell} > 0;\ \Delta G < 0$

C) $E_{cell} < 0;\ \Delta G^0 > 0$ D) $E_{cell} > 0;\ \Delta G^0 < 0$

ANSWER: B

18. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be

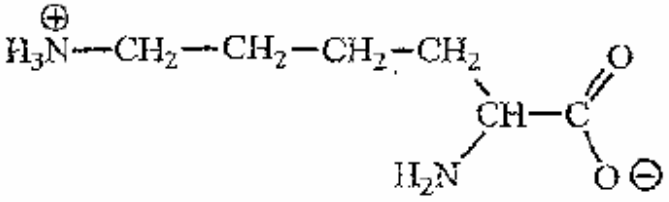
A) 35 mV B) 70 mV

C) 140 mV D) 700 mV

ANSWER: C

SECTION – IV (Integer Type)

19. The total number of basic groups in the following form of lysine is

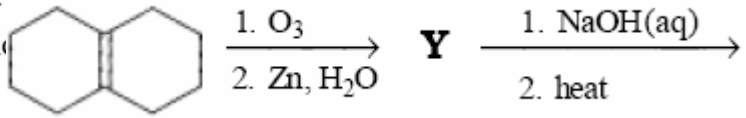


ANSWER: 2

20. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is

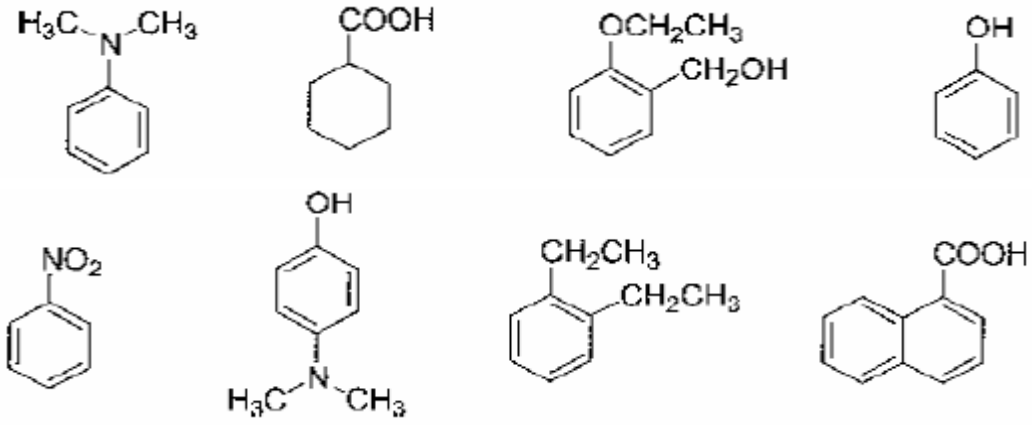
ANSWER: 5

21. In the ecular aldol condensation produ



ANSWER: 1

22. Amongst the following, the total number of compounds soluble in aqueous NaOH is



ANSWER: 4

23. Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is

KCN	K_2SO_4	$(NH_4)_2C_2O_4$	NaCl	$Zn(NO_3)_2$
$FeCl_3$	K_2CO_3	NH_4NO_3	LiCN	

ANSWER: 3

24. Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is

ANSWER: either 0 or 8

25. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is

ANSWER: 3

26. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titre value is

ANSWER: 3

27. The concentration of R in the reaction R $\rightarrow$ P was measured as a function of time and the following data is obtained:

[R] (molar)	1.0	0.75	0.40	0.10
t(min.)	0.0	0.05	0.12	0.18

The order of the reaction is

ANSWER: 0

28. The number of neutrons emitted when $^{235}_{92}U$ undergoes controlled nuclear fission to $^{142}_{54}Xe$ and $^{90}_{38}Sr$ is

ANSWER: 4